

Particle filter vs characteristic-function likelihood

Which non-Gaussian filter to adopt for variance at the zero boundary. A comparison for your decision. Session H, 15 September 2026.

The problem

Session G left one flag on the rough host. When variance sits exactly at zero on 10–17% of days ($\xi = 0.3$, $\theta = 0.04$, $\kappa = 3$), the exact-moment Kalman filter's Gaussian quasi-likelihood is unreliable. Matching two conditional moments is not enough when the law of tomorrow's variance has an atom at zero. Two non-Gaussian filters were built today and benchmarked on the same data.

	adapted particle filter (<code>filters/particle.py</code>)	characteristic-function filter (<code>filters/fourier.py</code>)
idea	simulate the state; weight by the observation	carry the conditional cf of V; Bayes' rule in Fourier space (Bates, 2006)
transition law	the QE step itself, with <code>substeps</code> per day	exact for the continuous lift: the variance cf solved as a Riccati system
observation step	fully adapted: each particle's predictive likelihood in closed form (atom included), resampled on it, V drawn from its exact posterior	exact Fourier integral for p(y); posterior mean and variance of V from the cf's derivatives
what is approximated	nothing but Monte Carlo (500 particles; the likelihood estimate is unbiased)	the posterior after each update: V gamma, the factors Gaussian given V
randomness	yes: common random numbers and sorted resampling make it reproducible and smooth in the parameters	none: a deterministic, smooth likelihood
observations	spot variance (adapted); daily realised variance in the bootstrap version	spot variance only (realised variance needs the joint cf of V and $\int V$)

Both were checked against independent references before the benchmark

(`test_nongaussian.py`):

- **Predictive densities.** Both integrate to 1 (0.99995 and 0.99969 near zero).
- **The cf's Riccati.** It matches the exact affine conditional mean and variance to 3e-8.
- **The QE draw.** The particle filter's noise reproduces the QE step draw for draw.

Along the way one numerical bug was found and fixed: an overflow in the particle filter's exponential-branch density when the conditional mean is tiny.

Results: 6 seeds near zero, 4 away, 1500 days each

κ was profiled on a grid with θ , ξ and R held at the truth (`study_zero_boundary.py`).

NEAR ZERO ($V = 0$ ON 10–17% OF DAYS)

	exact-moment Kalman	cf filter	adapted particle filter
mean κ (truth 3)	3.72	2.69	3.28
κ minus the particle filter's, per seed	-0.04, +1.53, +0.09, -0.03, +1.12, -0.02	-0.54, -0.38, -0.86, -0.70, -0.35, -0.70	0
log-likelihood at the truth, gain over Kalman	0	+256.5	+257.5
cf minus particle log- likelihood, per seed		-2.1, +4.4, -4.1, -1.1, -4.4, +2.0	
filtered-variance RMSE ($\times 10^{-3}$)	9.47	8.49	8.49
90% band coverage		0.91 (moment band)	0.86 (particle quantiles)
seconds per likelihood, machine under load	1.2	14	25
per day, measured back to back (throttled CPU; ratios comparable)	0.65 ms	2.6 ms + 0.8 s per parameter set	11.6 ms (500 particles, 4 substeps)

Kalman's κ misses are irregular here. On 4 of 6 seeds it sits within 0.1 of the particle filter; on the other 2 it is off by +1.1 and +1.5. On the finer-simulated data below it reads high on all three seeds. Either way it is not an estimator to use at the boundary.

Both non-Gaussian filters see the same data. They agree on the likelihood to within the particle filter's Monte Carlo noise, about 3.5 nats per 1500 days, and they filter V equally well: 10% lower RMSE than Kalman.

Yet the cf filter's $\hat{\kappa}$ is below the particle filter's on every seed, by 0.6 on average. That is systematic, not noise. The next section finds where it comes from.

AWAY FROM ZERO (V NEVER NEAR ZERO)

All three agree. $\hat{\kappa} = 2.61, 2.66$ and 2.62 . Log-likelihoods are within 4 nats of each other. RMSE matches to 0.03%. Where the Gaussian is right, the non-Gaussian filters cost 5–18× more for nothing.

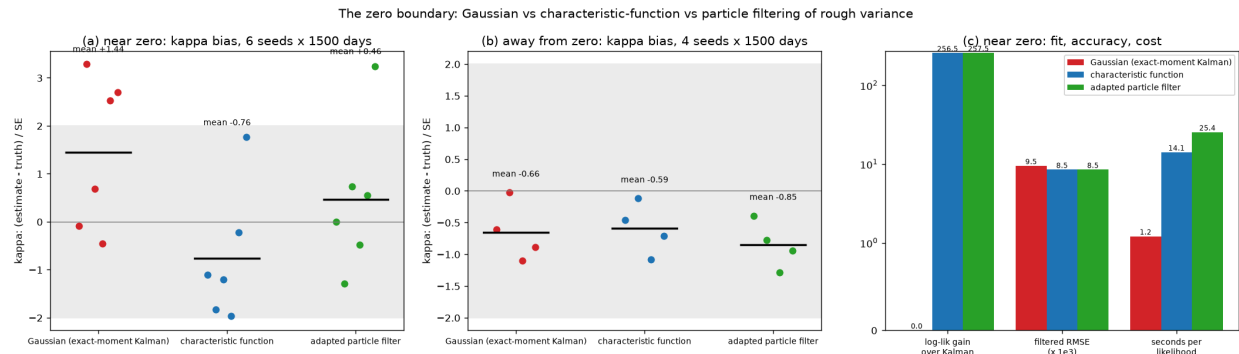
Where the cf filter's low κ comes from

The benchmark data were simulated with 4 QE substeps a day, and the particle filter runs on exactly that law. The cf filter models the continuous lift. Part of its offset could therefore be a model mismatch against discretised data, not its gamma approximation. To separate the two, the same study was repeated on data simulated with **64 substeps a day**, close to the continuous law, with the particle filter at 4 and at 16 substeps (`study_zero_boundary_fine.py`):

$\hat{\kappa}$ on 64-substep data	seed 61	seed 63	seed 65	mean	minus particle (16 substeps): mean $\pm$ SE
exact-moment Kalman	2.94	4.21	3.81	3.65	+1.05 $\pm$ 0.35
cf filter	2.39	2.19	2.82	2.47	-0.13 $\pm$ 0.20
particle filter, 4 substeps	2.63	2.90	3.28	2.94	+0.34 $\pm$ 0.18
particle filter, 16 substeps	2.55	2.65	2.60	2.60	0

The offset belonged to the data's law, not to the gamma approximation. Once the data follow the continuous lift, the cf filter's κ is within noise of the particle filter that simulates that law closely (16 substeps). Now the **4-substep particle filter** is the one that reads high. The particle filter is consistent for exactly the law it simulates, and at 4 substeps that is the QE discretisation, not the continuous model. The Kalman filter stays off by about +1 on this data.

Three seeds is a small sample: an offset below about 0.4 in κ cannot be excluded. But the picture from the main benchmark, a cf filter biased on every seed, did not survive the change of data. The first run on this data also exposed the particle filter's overflow bug: one seed's profile had come out with a zero standard error.



Trade-offs

	cf filter	adapted particle filter
likelihood surface	deterministic and smooth: quasi-Newton optimisers, Hessian standard errors	Monte Carlo: smooth enough to profile under common random numbers (second differences under 0.5 nats), not for Hessians
accuracy near zero	likelihood and filtering as good as the particle filter; κ within noise of it on continuous-law data	consistent for the law it simulates: enough substeps for continuous-time data
model it targets	the continuous lift	the QE-discretised lift; more substeps approach the continuous law at proportional cost
cost per likelihood, 1500 days	~5× Kalman: 2.6 ms a day plus 0.8 s of Riccati per parameter set	~18× Kalman at 500 particles and 4 substeps; linear in particles × substeps, so ~4× more at 16
realised variance	needs the joint cf of $(V, \sqrt{V})$: a known extension, not built	bootstrap version has it; the adapted step extends in closed form ($\sqrt{V}$ is linear in the last V)
finer lift ($N = 40$)	Riccati cost linear in N	per-step cost quadratic in N
uncertainty bands	from the moment-matched posterior	particle quantiles

Recommendation

For real data, whose law is continuous, make the cf filter the likelihood at the zero boundary. Keep the particle filter as its independent check.

1. **Default: the exact-moment Kalman filter.** It is right away from zero and 5–18× cheaper. Diagnose the boundary with it first: the share of days whose predicted variance is within two standard deviations of zero. Above about 5%, switch.

2. At the boundary: estimate with the cf filter.

- It targets the continuous lift, which is the model being fitted.
- Its likelihood is deterministic and smooth, so quasi-Newton optimisers and Hessian standard errors work.
- It matches the particle filter's likelihood and filtering accuracy.

3. Confirm with the adapted particle filter at 16 or more substeps around the cf filter's optimum: a profile in κ , and in H when that is learned. The two are independent derivations, and disagreement between them is the alarm. Four substeps is not enough for continuous-time data (+0.34 in κ above).

4. Two extensions before the history's realised variance goes through them:

- the cf filter needs the joint cf of the day's variance and its integral, which the same Riccati machinery gives with one more source term;
- the adapted particle filter needs the realised-variance observation in its last substep, which is closed-form because the day's integral is linear in the last draw.

Until then, the Kalman realised-variance filter stays the history's estimator, with the boundary diagnostic reported next to it.

Your call: confirm this policy, or prefer the particle filter as the primary estimator. That would mean choosing its substep count, and its cost, as a modelling parameter.